BRD: SMART-STOCK

Background

The hardware store lacks an efficient way to track sales and monitor stock. Staff have limited computer skills, so complex solutions won't work. The store needs an intuitive system that forecasts demand to prevent stockouts and protects sensitive data with strict role-based access.

Purpose and Scope

SMART-STOCK is an inventory management system designed for a hardware store. Logged-in users can process sales, monitor real-time stock, receive automatic restocking suggestions based on demand, and track which user handled each transaction. The system enforces role-based access control (Admin vs. Staff) and uses a simplified GUI tailored for nontechnical users.

High-Level Features

- Account Management (Admin and Staff)
- Simplified Point of Sale (POS)
- Inventory Monitoring
- Demand Forecasting & Order Suggestions
- Transaction Tracking

Limitations

- Customer features: No customer-facing features for storefronts and e-commerce.
- Multi-branch syncing: Designed for a single physical hardware store location.
- Accounting/HR: No complex financial ledgers, payroll, or employee time-tracking.
- Hardware integrations: No dedicated barcode scanner or receipt printer integrations in the initial phase (web/keyboard input only).

Stakeholders

- Store Owner (Client / Admin)
- Hardware Store Staff / Cashiers (Users)
- Yves Carranza (BA / QA)
- Indaleen Quinsayas (Approver)

Functional Requirements

Requirement	Priority	Source
The system shall provide a highly simplified, distraction-free GUI for processing sales.	Must	Client
The system shall allow Admin accounts full access to view overall inventory levels.	Must	Client
The system shall allow a user to log in with an assigned role (Admin or Staff).	Must	Yves
The system shall restrict Staff accounts to the sales interface only.	Must	Yves
The system shall record the name of the user/cashier handling each transaction.	Must	Yves
The system shall generate automated restocking suggestions for Admins based on sales demand.	Must	Yves
The system shall allow Admins to view the transaction history and the staff member who processed them.	Must	Client

The system shall automatically deduct stock from the inventory when a sale is finalized.	Must	Yves
The system shall create backups regularly.	Should	Yves
The system shall create a transaction summary.	Should	Yves
The summary shall be downloadable.	Should	Yves

Non-Functional Requirements

Requirement	Category	Priority	Source
The sales interface must be intuitive enough for non-technical staff to operate with minimal training.	Usability	Must	Client
The system should be operable on older/slower devices	Usability	Should	Client

Inventory deductions must be reflected in the database immediately after a sale completes to prevent overselling.	Performance	Must	Dan - SA
Staff accounts shall be unable to view or modify Admin-level data.	Security	Must	Yves
Passwords shall be securely hashed and never stored in plain text.	Security	Must	Yves
The system shall process each transaction in under 1 second	Performance	Must	Yves

Open Questions

Question	Owner	Status
How far back should the system analyze sales data to calculate the demand for order suggestions?	Yves	Tracked in SS-35

Does the client need a way to manually adjust stock levels for lost or damaged items, outside of making a sale?	Dan - SA	Resolved — see SS-38
Do we need to input the cost price of items, or just the selling price?	Ryza - Dev	Open
How many users should the system handle?	Vincent - PM	Open

Definition of Terms

Term	Description
Admin	A user role with full system access, typically the store owner. Can view all data and suggestions.
Staff	A restricted user role meant only for cashiers. Access is limited to the simplified sales GUI.
Order Suggestion	An automated recommendation generated by the system detailing what items to restock based on how fast they are selling.
Transaction Record	A log of a completed sale, including the items sold, total price, timestamp, and the staff member who handled it.